

Tribological characteristics of cast polyamide 6 (PA6G) matrix and their composite (PA6G SL) under normal and overload conditions using dynamic pin-on-plate system

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ABSTRACT

The current study addresses the tribological characteristics of Cast Polyamide 6 (PA 6G) matrix and its composite filled with polyethylene (PE) solid lubricant (PA 6G SL). Tribological tests under different contact kinematics (1) traditional pin-on-disc (PoD) and (2) dynamic pin-on-plate (PoP). Tribotests revealed that the PA 6G SL composite showed increased wear resistance and reduction in heat generation as compared to the virgin PA 6G matrix. The difference in tribological characteristics was analyzed with respect to morphology (primary and secondary layer) of the transfer layer formed on the counter material. In summary, the solid lubricant alters the tribological process by means of altering the surface response by introducing a limitation in the wear particle generation.

1. Introduction

In the recent years, polymer composites have gained significant importance as bearing material due to the properties such as lightweight, corrosion resistance, processing flexibility and most importantly self-lubricating nature [1–3]. Already, several engineered polymers are used as bearing materials in demanding engineering applications such as wind turbines, marine bearings and chemical reactors [4,5]. In these applications, the bearing is frequently subjected to severe loading conditions and supposed to have long-term sustainable performance. In order to facilitate the range of operating conditions for eg., expanding the loading regime. It is required to enhance the strength by introducing additives to the matrix. Besides mechanical strength, the tribological properties are to be controlled by the careful choice of tribo-additives.

Amongst several engineering polymers, Polyamides are considered as outstanding semi-finished engineering material for wear borne applications. Their tribological performance is frequently superior to POM, PET or PEEK due to good sliding and mechanical properties. The advantage of synergetic effects of the multiple properties is earlier studied for polyamides [6,7]. Given the consideration for tribological properties, cast polyamides are added with solid lubricants such as

PTFE, Molybdenum disulphide, wax, carbon fiber and graphite as filler materials for enhancement of frictional properties [8–10]. The current requirement in the industrial environment for lower density solutions proposes polyethylene as a potential filler material and few marketable grades of those are already existing. Available reports on the tribological behaviour of polyamide 6 (PA 6) – regardless of extruded, injection moulded or cast version - indicate the potential use of its composites in tribological applications. Recently Neis et al. [11] compared the tribological behaviour of neat cast polyamide 6 (PA 6G), PA6 G filled with molybdenum disulfide and PA6 G filled with a special solid lubricant to a commercially available bronze alloy. Highest wear rate was found in bronze alloy. PA 6G filled with a special solid lubricant showed low coefficient of friction (CoF) and superior wear resistance. In another research on PA 6 composites, additives such as talc, glass fiber, graphite show increased wear rate and CoF for talc and glass fiber [12]. This research also attempted a combination of solid lubricants (graphite and UHMWPE) which increased the CoF and wear rate of PA6. It is clear from existing reports that wax filled polyamide 6 showed low CoF when compared with the pure PA6 [13]. The comparative tribological study between PA 6 filled with liquid crystals and MoS₂ filled PA 6 clearly indicates the advantage of MoS₂ in reducing the friction, similar to that it was found for PEEK matrix [14,15].

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Bahadur and Palabiyik [16] studied mechanical and tribological properties of PA 6 and high density polyethylene (HDPE) polyblends. They found a decrease in the friction coefficient with increasing PA 6 content in HDPE. Addition of solid lubricants contributes to reducing the friction through the transfer layer formation. Since, the addition of lubricant can lead to the possibility of cohesive adhesion between the polymer and the transfer layer and reduces the friction. Many researchers [17–19] have made an extensive study of the transfer layer and solid lubricants. The performance of composites with solid lubricants under the large-scale testing is better when compared with the small-scale testing due to their larger contact surface area. The thin transfer layer formation in the small scale testing showed a favourable result in friction characteristics. Cast PA 6 has already seen applications in off-shore design, large-scale engineering constructions and harbour applications, where overloading may be anticipated.

Cast PA 6 G high crystalline polyamide is a popular industrial polymer for numerous light and heavy-duty applications. This material possesses a high hardness and dimensional stability with excellent mechanical properties and can be used as a wear resistant material. However, the tribological performance of PA 6 G matrix reinforced with the fillers has not been evaluated yet. Moreover, there is no overload and dynamic sliding condition results of PA 6 G and PA 6 G/PE composites in the literature. Besides the transfer layer characteristics and its effects on altering the tribological nature were scarcely reported in the chosen composite. In order to fill this research gap, the present research work was undertaken to address the tribological parameters of the matrix (PA6G) and its composites (PA6G/SL) in various loading and speed conditions.

2. Experiments and methods

2.1. Materials used

Cast polyamide 6 (PA 6 G) matrix and the composite filled with PE solid lubricant (PA 6 G SL) resulted in the production of semi-finished plate by Ensinger GmbH, supplied by Quattroplast Ltd, Hungary. Commercial brand names: Docamid 6G and Docaglide. Tensile strength 83–75 MPa, E modulus 3500–3400 MPa, Ball indentation hardness 175–170 MPa, E/H ratio 20–20 consequently.

2.2. Experimental measurement details

2.2.1. Tribological testing setup

Pin-on-disc measurements (Fig. 1) were carried out according to the wear test category VI of the German standard DIN 50322. Plastic samples with a contact diameter of 5 mm and a free height of 3 mm, were used as pins. S235 steel counterface (average roughness = $R_a = 0.06\text{--}0.1\ \mu\text{m}$) was slid against the static pins. The pin was positioned on the disc at a radius of 17 and 27 mm. During the measurements, the following parameters were simultaneously measured: the coefficient of friction (μ), based on the measured forces (F_x and F_y by full bridge strain gauges mounted on the specimen holder), the wear (characterized by the vertical displacement of the pin holder (d)) and the bulk temperature with the thermocouple at the surface vicinity (at

1 mm depth from the surface in the pin - of contact (T)). On the contact spot, the force of friction (F_s), parallel to the plane of the disc and the normal force by dead weight (F_n), perpendicular to the plane of the disc (F_z), is acting on the disc. The force of friction is the resultant of two forces (F_x and F_y), both parallel to the plane of the disc. The coefficient of friction (μ) can be calculated by dividing the force of friction by the normal force:

$$F_s = (F_x^2 + F_y^2)^{1/2} \text{ [N]}, \quad (1)$$

$$F_n = F_z \text{ [N]}, \quad (2)$$

$$\mu = F_s / F_n = (F_x^2 + F_y^2)^{1/2} / F_z \quad (3)$$

For the clear understanding, “Dynamic I.” and “dynamic II.” abbreviation of the test conditions are used to following the Wöhler cases logic for fatigue condition.

- Dynamic I. Condition: used for characterization of the pin-on-disc (PoD) system with constant sliding velocity, constant load and steady ambient laboratory conditions
- Dynamic II. Condition: used for characterization of the pin-on-plate (PoP) system with a programmed closed loop sliding path having a combined straight and curved wear track within a single cycle. Such a track produces an effect from acceleration and inertia force as a consequence of the curvature. Moreover variable PV values under steady ambient laboratory conditions were also studied.

Thus, PoD (dynamic I. Conditions) and PoP (dynamic II. Conditions) sliding coefficients of friction were also calculated to enable measurement of its frictional capability, wear and contact temperature at different PV regimes. The wettability and surface energy of both samples was also studied for the analysis of the contact mechanism between polymer and counter plate. Microscopic analysis was performed on the worn out surfaces of PA 6 G SL composite and counter plates to get an understanding of the formation of the transfer layer and debris effect on the wear track.

Beside the PoD test (dynamic I conditions) method, a specially developed PoP test (dynamic II conditions) system was also applied for tribological characterization of the selected material grades. It is generally known from the engineering experience that before reaching the steady state sliding (dynamic balance of polymer transfer layer) the running-in period is critical for the evolution of polymer transfer film and sensitive for the external dynamic load and speed effects. That is why the PoP test system was implemented to monitor and compare the running-in friction behaviour of the materials. The curvature of the wear track tends to explain the influence of the velocity gradient on the corners. Besides, such a pattern (See Fig. 2) has a tendency to achieve a maximum number of curved paths and straights in one single cycle. And to have similar velocity gradient across all the bends in a single cycle.

In the conventional PoD system, two different pv levels (compressive stress x sliding velocity) were investigated. The first is the “Low pv as LPV” which was typical for the normal application of plastic machine elements having a reasonable safety factor. The second is “High pv as HPV (overload case) for neat cast PA 6 G. Details of the tribological testing conditions are provided in Table 1. During the PoP friction tests (dynamic II. conditions), a plastic sample pin with a diameter of 5 mm were slid against the counterface along a programmed path (Fig. 2) with a dynamic change of test conditions under the control. The sliding speed was controlled in the curvature of the wear track so that the central point of the pin contact area has the same velocity as that of the straight path “v”. In the curved wear track, the speed distribution of the contacting pin can be described by the speed gradient that results dynamic stress built-up, which eventually diminishes after the curvature. Furthermore, the speed is combined with the inertial force effects at the curvature. The PoP test completed the full path for ten cycles at 5×2 average speed, which corresponds to 5 different pv regimes. In our

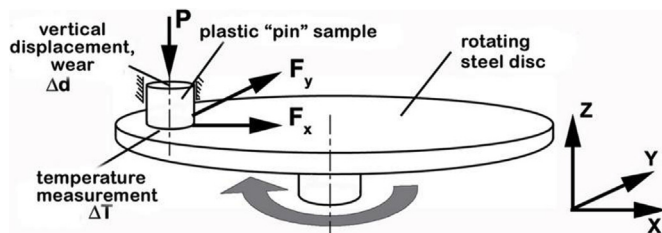


Fig. 1. Pin-on-disc setup for steady-state friction measurements.

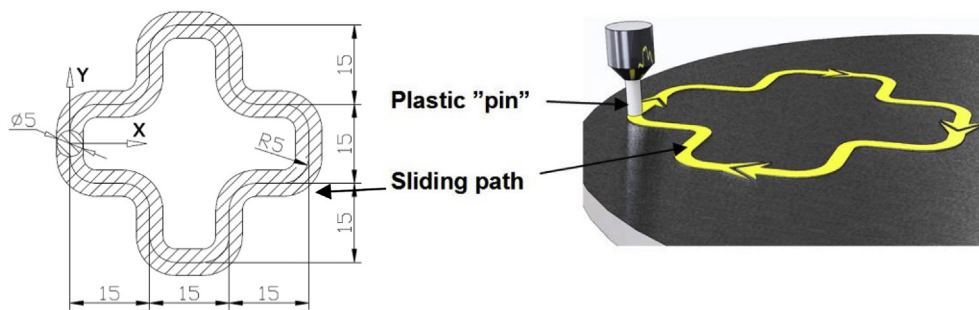


Fig. 2. Programmed dynamic sliding path of plastic samples on the fixed steel plate.
 - 1 cycle is composed by 8 straights and 12 bends forming one closed loop.
 - parallel, transition and perpendicular sliding to grinded grooves of steel surface.

tests, the normal load was constant (Table 1). During the evaluation, each speed level, which means 2 repeated cycles, was studied. According to the PoD measurements the same features were measured (F_x , F_y , d , T) and calculated (F_s) by the same instrumentation mounted on the dynamic module of the tribotester.

2.3. Surface characterization

Contact angle measurements were performed using the static sessile drop method and the SEE System apparatus (Advex Instruments, Czech Republic). Double distilled water and diiodomethane, DIM (Sigma-Aldrich, Reagent Plus 99% grade) were implemented as test liquids. A Hamilton syringe was used only for injecting 2 μ l droplets. The measurements were performed at 23 °C. The method of Owens and Wendt [20] was used for the calculation of the total surface energy and its polar and dispersive components. Microscopic analysis was made using the Zeiss Axio Lab A1 light microscope under various magnifications (100 $\times$ to 1000 $\times$) for the purpose of understanding the morphological analysis of the worn surfaces.

3. Results and discussions

3.1. Wettability and surface energy

Friction is highly influenced by the adhesion of contacting surfaces, which are often characterized by the surface energy, hence the different material grades were characterized for their surface energy. Fig. 3 depicts the sessile drops of water and diiodomethane in the neat cast PA6G and the composite PA6 G SL samples. The measurements were repeated twelve times. The highest and lowest values were omitted and 10 values were used for further calculations. Table 2 shows the corresponding values of contact angles in full ranges, and the calculated average values of the total surface free energy (γ_{tot}) with its relative ($\sigma_{\%}$) and standard deviations (σ) and the calculated average polar (γ_{polar}) and dispersive (γ_{disp}) components. The composite compared to the neat one shows an increase in the contact angles, thus the total

surface energy, and its polar and dispersive components decreased moderately. While the difference in the averaged values of surface energy was about 5%, the relative standard deviations of total surface energy was 2–3%. However, a relatively small decrease in surface energy of the composites refers to the effect of reinforcing materials, which could influence (reduce) the adhesion component during sliding [21] and it will be discussed in the following section.

3.2. Friction behaviour on PoP system

In regards to the PoP system, the sliding path was programmed to a specific pattern. During the beginning of the experiment, the influence of the directionality of the surface micro-topography was investigated. This was done by performing individual tests by engaging the sample pin to move parallel as well as normal to the direction of the grooves in the counter plate. Hence two sets of experiments in PA 6 G/1 and PA 6 G/2 were performed in accordance with the initial starting position.

The PoP system was developed to investigate any possible difference in sensitivity against dynamic conditions during running-in friction of the two PA6 G types. As described in 2.2.1 based on our preliminary tests 5 $\times$ 2 repeated cycles (Fig. 2) having 5 pv regimes were sufficient to produce the running-in transfer film. The friction characteristic curves of both neat polymer and its composites with five different levels of speed are shown in Fig. 4. Two repeated measurements are presented for each case, but the samples were at different starting positions compared to the grinded grooves of steel surface (parallel and perpendicular). The results of different starting positions can be followed by the friction curves, where friction force and coefficients vary at a reverse sign. In order to map the friction behaviour for a short period, running-in dynamic tests were performed. The short test duration avoids the uncertainties that may arise from wear, deformation, heat generation and the dynamically involved in the transfer layer characteristics such as peeling and replenishing of transfer layer and the morphological differences between the transfer layer can be eliminated in the short duration test. The measured friction of polymer samples placed into different starting positions to ground grooves of steel is

Table 1
Summary of tribological testing conditions.

	Pin-on-disc, PoD (LPV) (dynamic I_L conditions) ($p_v = 0.43 \text{ MPa ms}^{-1}$)	Pin-on-disc, PoD (HPV) (dynamic I_L conditions) ($p_v = 1.79 \text{ MPa ms}^{-1}$)	Pin-on-plate, PoP (dynamic I_L conditions) (pv regime 0.024–0.24 MPa ms^{-1})
Sliding speed [m/s]	0.14	0.22	0.008–0.08
Period of test [h]	4	4	10 (5 $\times$ 2) cycles
Surface load [MPa]	3.05	8.14	3.5 + inertia
Friction path radius [mm]	17	27	Dynamic program path
Disk surface roughness [μm]	R_a 0.06–0.1	R_a 0.06–0.1	R_a 0.06–0.1
Ambient temperature [°C]	23	23	23
Ambient humidity RH [%]	57	57	57

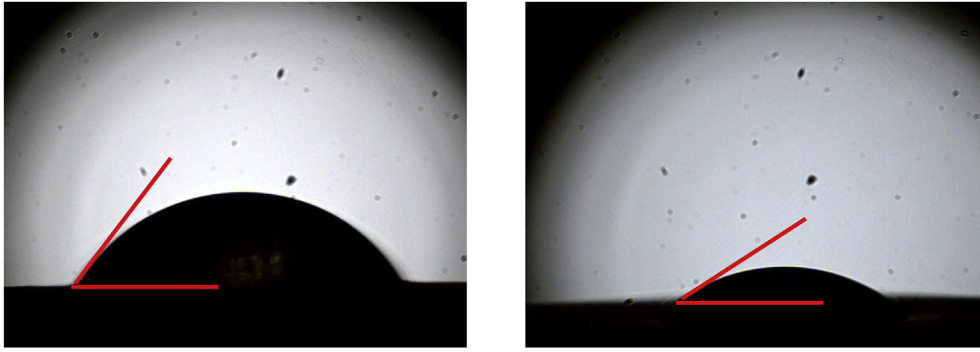


Fig. 3. Contact angles on PA6G and PA6G SL (Left: water, right: diiodomethane, top: Polymer (PA6-G), bottom: (PA-6G SL composite)).

Table 2

Contact angles, surface energy and its components for the matrix PA 6G and PA6 G SL composite.

Sample	θ_w deg	θ_{DIM} deg	γ_{tot} mJ m ⁻²	σ of γ_{tot} mJ m ⁻²	$\sigma_{\%}$ of γ_{tot}	γ_{polar} mJ m ⁻²	γ_{disp} mJ m ⁻²
Composite	55.1 ± 2.2	28.9 ± 2.4	57.9	1.85	3.2	13.2	44.7
Natural	52.2 ± 2.9	27.7 ± 2.6	59.8	2.57	4.3	14.6	45.2

shown in Fig. 4a (PA6G/1, PA6G/2 and PA6G SL/1, PA6G SL/2 respectively). The effects of inertia forces during the bends and increasing speed level on friction coefficients can be followed. The amplitude of the friction force was higher (friction coefficients were between 0.2 and 0.32) during the lower speed range (cycle 1–4) owing to the deformation of micro-geometry at the contact zone crossing the grinded grooves. Moreover, during the initial period contact, a non-equilibrium state prevails due to the difference in the micro-topography between both the contact surfaces. Such a phenomenon is overcome by the formation of the transfer layer to achieve the equilibrium between the two surfaces by having a smooth uniform profile. The impact of parallel and perpendicular starting positions to steel grooves was clear at cycles 1–4, where alternatively reversing trends of the friction curve are seen from the green and black curves (see Fig. 5a). With the increasing speed level, time dependent strain behaviour of the polymer plays an important part. Limited time for the deformation component of friction results in reducing the friction with a smaller amplitude of the friction force within each dynamic cycle (9–10 vary between 0.22 and 0.27). The averaged friction coefficient from multiple cycles is shown in Table 3, Fig. 4b shows the results of the composite measurements in the same manner as it was shown for the neat polymers.

From the point of engineering applications, no significant difference in the friction behaviour was found between neat polymers and composite samples. At low speed, the internal lubricant and lower surface energy with a generally accepted normal load do not work in the case of composite samples. Considerable variations in friction coefficient within a dynamic cycle can be monitored (range of 0.18–0.33). No audible sound (friction instability or stick-slip) was evident with neat polymers, on the contrary, the friction sound was found in the composite. So, low speed and stick-slip behaviour or friction instability of this type of composite material needs further investigation.

3.3. Friction, wear and temperature behaviour of polymer and composite at low pv (LPV)

The friction, wear and temperature characteristic profiles of neat polymer and its composites measured in the PoD system are illustrated in Fig. 5. Two repeated measurements in each case have been plotted. In the figure, the codes PA6G-1 and PA6G-2 and PA6G SL1 and PA6G SL2 indicate the two neat polymers and composites, respectively.

Fig. 5a shows in the absence of any significant difference in friction characteristics between neat polymers and composite for the normal (but well loaded) sliding application results. The self-lubricating samples resulted in relatively lower friction coefficients, moderate wear (Fig. 5b) and deformation with less heat generation (Fig. 5c). This behaviour refers to the lower surface energy and low adhesion of the composite samples. According to the Archard's friction theory [22], the friction force is the sum of the adhesive and deformation components of friction – the deformation at the applied low load level with similar strength and hardness values already introduced in 2.1. may not differ significantly. Thus, as a result of low adhesion, a reduced friction was observed for the composite. It is interesting to note that the friction coefficients are stabilized around 0.48–0.5 in the final stage of the testing procedure. This behaviour is already known for composites: when the load and wear are moderate, the lubricant materials do not have the ability to reach the surface in sufficient quantity resulting little effects on the friction and wear modification.

In regards to the tribological signals (temperature, friction and wear) all three of them follow a similar pattern. Apart from the similarity between the three signals, there is likewise a similarity in the running-in behaviour between the temperature and the friction signals. The overall trend in the friction and temperature curve shows a running-in phase followed by an intermediate transition state which subsequently leads to a steady-state. As observed from the plots (Fig. 5) the sliding distance required to achieve the transition zone varies with respect to the material in the following order (increasing) PA 6 G/1, PA 6 G/2, PA 6 G SL/1, PA 6 G SL/2. The plots are a clear indication that the sliding distance for achieving the steady-state is relatively less for the neat polymers. This is due to the formation of a transfer layer, which is much more prone for neat polymers as also seen from the wear curve (Fig. 5b).

One of the main reasons for the different stages in friction and temperature curves is due to the irregularity in the contact surface from the initial micro-topography. A smooth uniform surface is required (for both polymer and the counterface) to attain the equilibrium for achieving the steady state. On comparing both the materials, the neat polymers have shown more wear as seen from Fig. 5b. These generated wear particles contribute to the formation of the transfer layer. Hence with the increased level of the generated wear particle, the transition state is reached at a relatively shorter sliding distance. In composites, it is evident that the wear is relatively lower than the neat polymers, which can be attributed to the lubrication effects. Hence a relatively substantial time is needed to generate particles for the full formation of a transfer layer. The prolonged running-in signal of the composites supports the above. A relatively low temperature in the composite material can be attributed to the lesser frictional heating as a consequence of the solid lubricant. The difference in sliding distance required to achieve the steady state between the neat polymer and composites is justified by the sliding distance required to form a sufficient transfer layer. However, this requires further research with *in-situ*

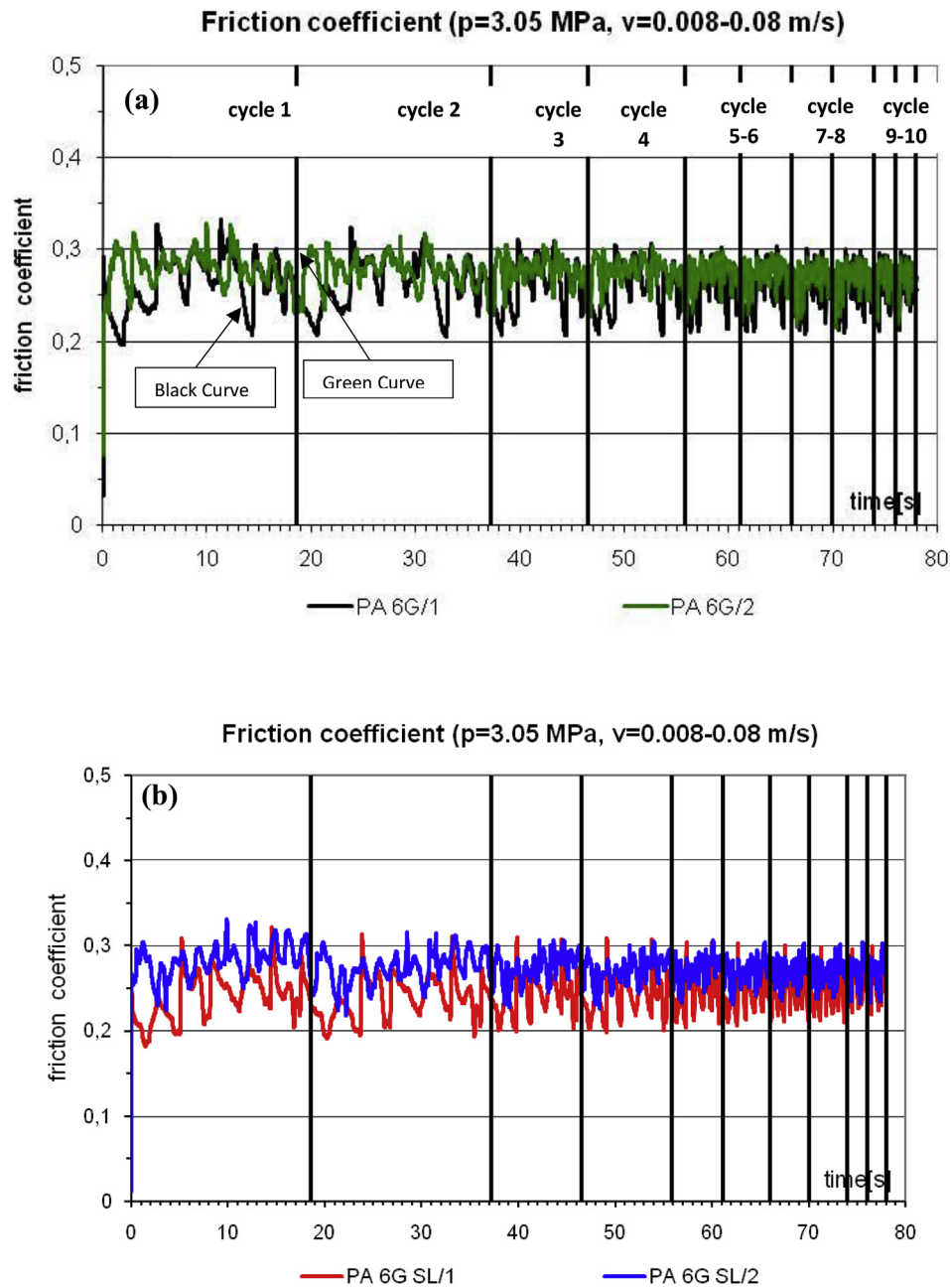


Fig. 4. Friction coefficients of natural cast PA6 samples (a) composite PA6 (b) under 10 cycles, 5 speed levels ($v_{1-2} = 0.0083$ m/s, $v_{3-4} = 0.016$ m/s, $v_{5-6} = 0.03$ m/s, $v_{7-8} = 0.041$ m/s, $v_{9-10} = 0.083$ m/s).

monitoring of the transfer layer using the vision monitoring system.

Fig. 5a clearly shows the peculiar frictional characteristics of neat polymer with its local maximum at the transition zone between the running-in and the steady-state. However, this is not the situation with the composite where a smooth transition was observed. This phenomenon in neat polymer can be explained from the viewpoint of increased deposition of wear particles on the counterface. On the other hand, the composite having a relatively lower wear rate produces less weaker transfer layer due to the effect of lubricants. During the initial phase of normal operating conditions (LPV) a slight difference in friction and wear behaviour was realized between the neat polymer and its composite. The self-lubricating effect could be realized after a min and 50 m sliding distance (Fig. 5a). During the steady state period, almost a constant trend was observed in the case of the neat polymer, whereas, a significant fluctuation in the friction curve was seen in the case of composites especially between 500 m and 1500 m. This could be due to

the mobility of solid lubricant between the composite and counter plate. On the other hand, the friction curve of both polymer and composite was considered to be nearly merged at the end of the test. At the same time, the linear increase (Fig. 5b) in the wear loss was observed in both cases which indicates the removal of material because of mechanical loading rather than the temperature effects. Constant profile of contact temperature at steady-state condition (Fig. 5c) indicates the formation of transfer layer which could provide a mask over the valley in the counter plate. A detailed explanation of the transfer layer characteristics is given in the subsequent section.

3.4. Friction, wear and temperature behaviour of polymer and composite at high pv (HPV)

The previous section shows the absence of significant difference in the friction value between polymer and composites in the specified LPV

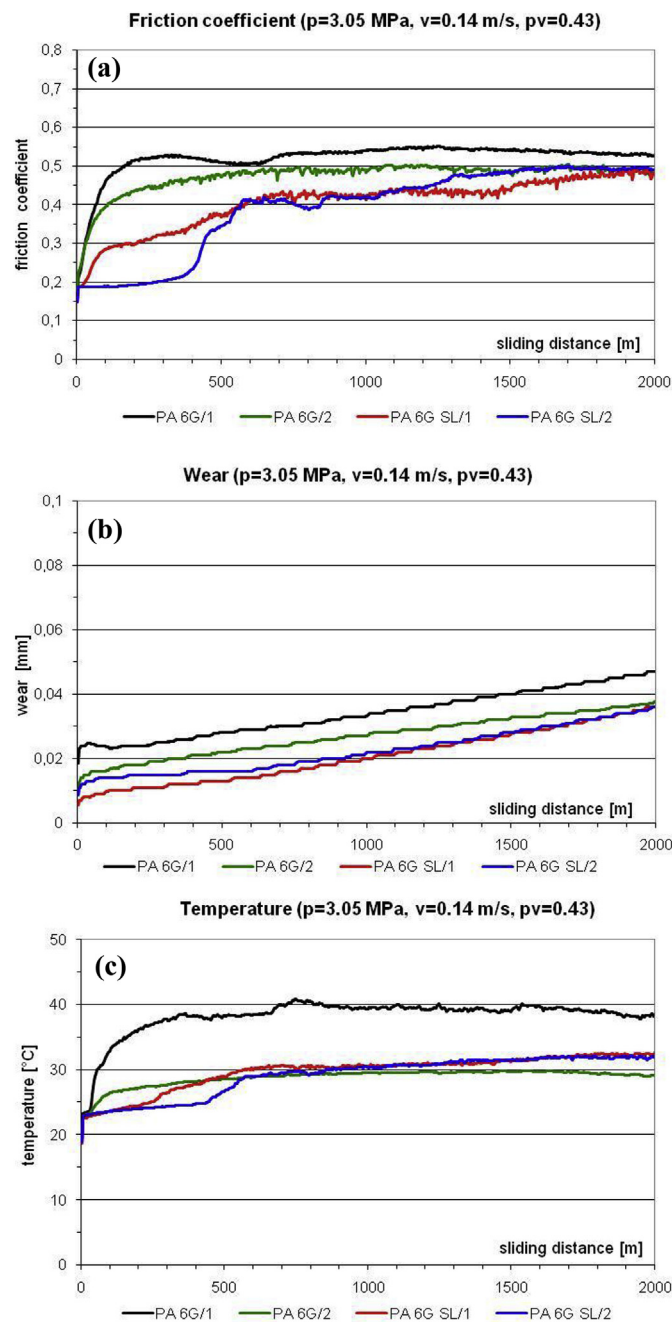


Fig. 5. Pin-on-disc model results of normal operating condition of a polymer slide bearing: $p_v = 0.43 \text{ MPa ms}^{-1}$ with $v = 0.14 \text{ m/s}$, dry friction.

Table 3

Averaged friction coefficients for different testing cycles.

Speed level	Cycle	PA 6G/1	PA 6G/2	PA 6G SL/1	PA 6G SL/2
μ_{av}					
0.008	1	0.265	0.279	0.241	0.283
	2	0.264	0.277	0.239	0.274
0.016	3	0.263	0.277	0.242	0.274
	4	0.262	0.276	0.241	0.271
0.03	5	0.261	0.273	0.244	0.272
	6	0.260	0.273	0.243	0.270
0.041	7	0.260	0.271	0.245	0.270
	8	0.260	0.271	0.243	0.690
0.083	9	0.261	0.269	0.251	0.690
	10	0.260	0.268	0.249	0.680

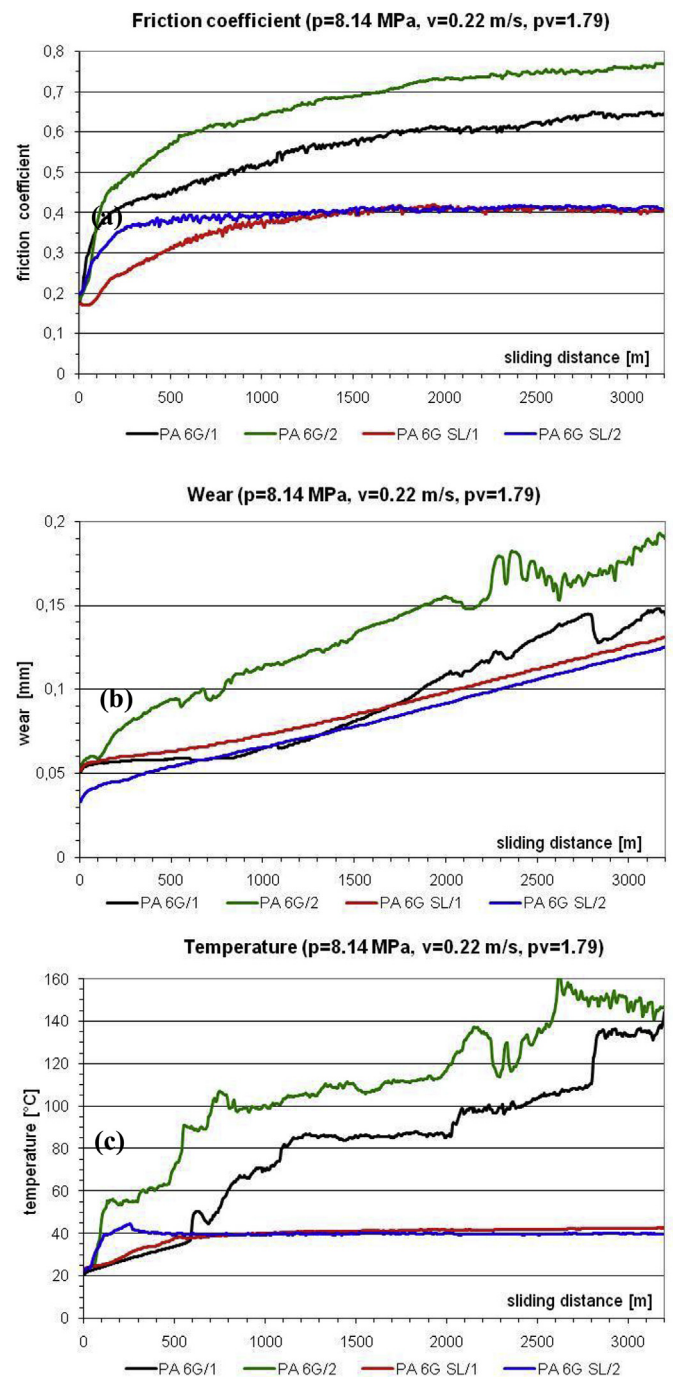


Fig. 6. Friction, wear and heat generation in pin-on-disc model of overload operating condition. $p_v = 1.79 \text{ MPa ms}^{-1}$ with $v = 0.22 \text{ m/s}$, dry friction.

regime. Hence, both polymer and composites were subjected to the tribological testing for HPV. The friction, wear and temperature curves are given in Fig. 6. It is clear that the resulting friction coefficients from HPV are lower than that of the LPV settings. This validates the influence of load on the self-lubricating ability of composite materials. The wear process (Fig. 6b) supports the action of lubricating the tribological pair at the contact interface. The resulting adhesion and deformation of plastic in the contact zone can be neat polymers samples reaching very high values (app. 0.8) with the melting process of the plastic surfaces (see the microscopic photos later). The melting phenomenon can be applied in both the wear and temperature curves (Fig. 6b and c). After 2000 m sliding, the modified contact gave loud low-frequency noise. The dynamic balance between adhesion and re-adhesion of PA 6 film on

Table 4

Early stage of friction for the matrix PA 6G and PA6G SL composite (Indicative values).

	$\mu_{\max}$	μ_{av}
PA 6G/1	0.2	0.21
PA 6G/2	0.23	0.22
PA 6G SL1	0.18	0.18
PA 6G SL2	0.19	0.18

the steel surface is not valid anymore. The measured bulk temperature (close to the contact, but not in contact) reached 180 °C, which can also assume the softening and melting in the contact zone.

Absence of the melting process was realized for the composite under

the HPV condition. Continuous but even wear and stabilized friction heat generation were recorded (Fig. 6c). The maximum limit for the bulk temperature was constrained at 40 °C. During the steady state condition, a constant friction coefficient of 0.4 value was noticed which could be due to the real dynamic balance of polymer film adhesion on the steel disc surface in case of composites. Compared to a normal test condition, the absence of any significant differences between $\mu_{\max}$ and primary μ_{av} is a consequence of increased deformation component of friction as seen in Table 4.

In regards to the pattern of three tribological signals, the HPV has a tendency similar to that of LPV in the order of ranking between different tests. The sliding distance required to achieve the transition zone is given in an increasing order for the following tests (PA 6 G/1, PA 6 G/2, PA 6 G SL/1 and PA 6 G SL/2). In regards to the friction

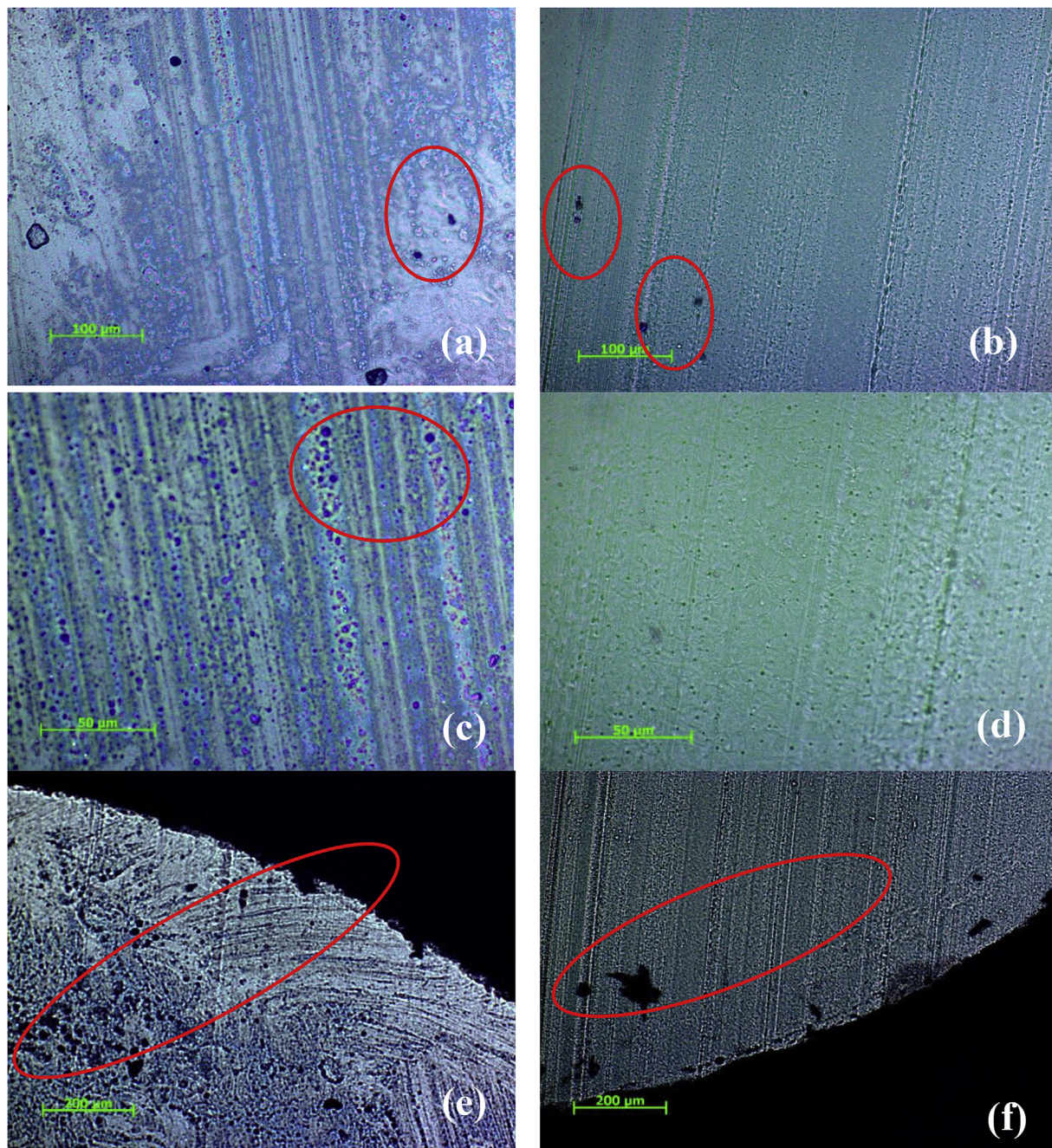


Fig. 7. Microscopic topography analysis of the samples obtained at $p_v = 0.43 \text{ MPa ms}^{-1}$ with $v = 0.14 \text{ m/s}$, dry friction condition: (a, c, and e) worn surfaces of natural polymer; (b,d,f) worn surfaces of composites.

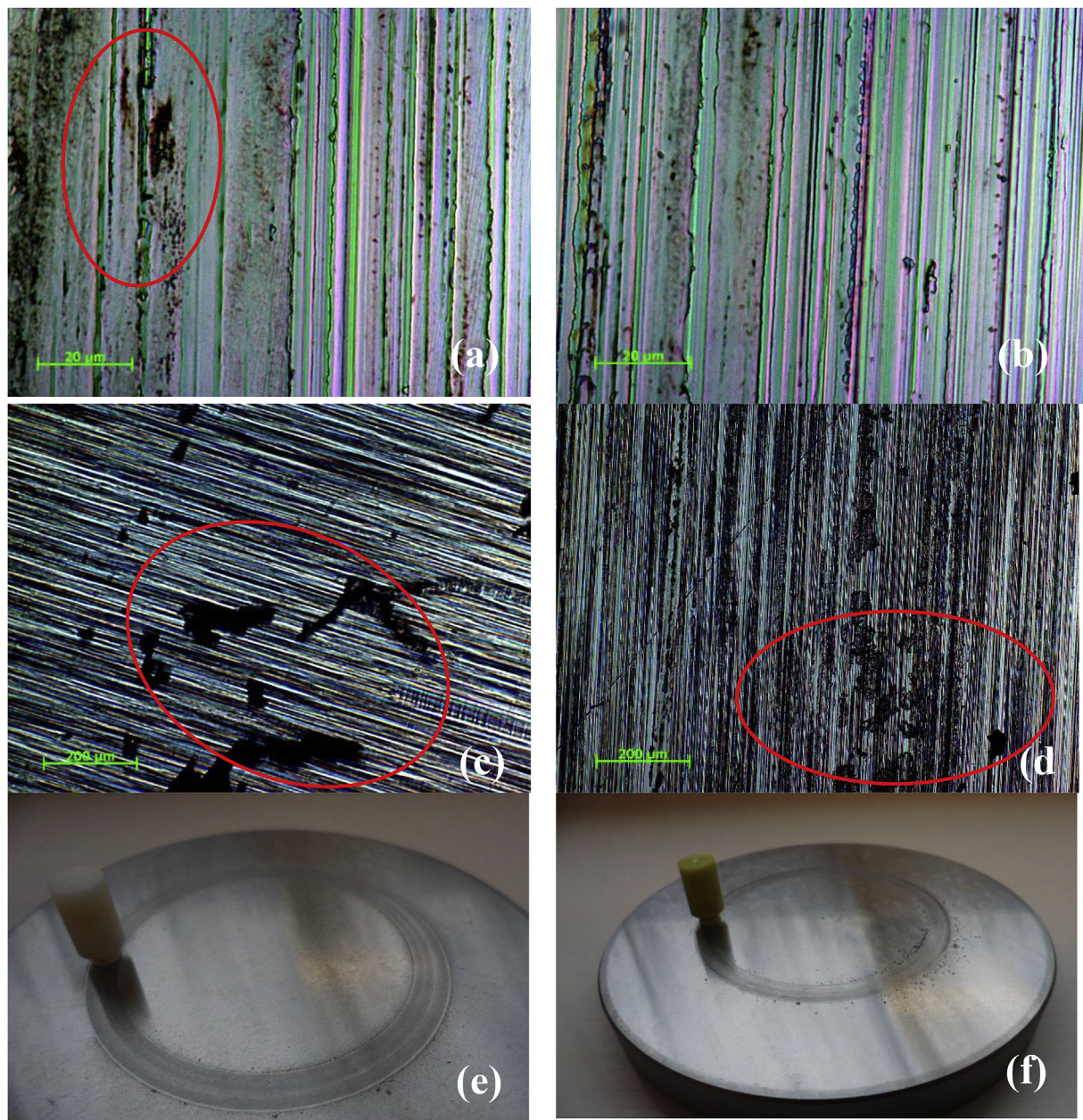


Fig. 8. Microscopic topography analysis of the counter plate obtained at $p_v = 0.43 \text{ MPa ms}^{-1}$ with $v = 0.14 \text{ m/s}$, dry friction condition: (a, c, and e) against with natural polymer; (b,d,f) against with composites.

characteristics, both the tests PA 6 G/1 and PA 6 G/2 reached a transition state. This also means that for the same sliding distance as that of the composites, a steady state is not obtained even at the last cycle of testing for the neat polymer. The difference in the sliding distance between the two materials to reach the transition and also the steady state can be ascribed to the formation of the transfer layer. Fig. 6b clearly stresses that the increased wear for neat material allows a fast pace to move towards the transition period.

Beside the friction curve, the wear and temperature profiles show interesting phenomena of the peeling-off of the transfer layer for the neat polymer. In the wear curve, a drop in wear was noticed around 2000 m from the wear profile. This can be ascribed to the removal of the transfer layer. The temperature profile in the identical location appears to increase. This is apparently due to the fact that the hindrance caused by the transfer layer towards thermal dissipation from the steel was cleared by the peeling of the transfer layer. Hence there is an increase in temperature in a region close to 2000 m sliding distance for

neat polymers.

4. Microscopic analysis

4.1. Worn surface of samples and counter plate at LPV

Fig. 7 shows the worn surface of the neat polymer following the wear tests in LPV limit. The topography, of the polymer (Fig. 7a) is expected to results in the observation of a severe deformation close to the edge and more re-adhered particles in the middle. It indicates higher stress concentration at the edges as the possible reason for creating this damage at the tested level of PV value for the neat PA 6. This picture shows partial damage referring the exceeded PV limit of the material. Reports of a similar stress concentration finding are provided in the literature [23]. The PV limit is a key factor which can cause damage to the worn pin from a moderate to large scale as reported in the literature [24]. Further, micrograph analysis of PA 6 (Fig. 7c) at

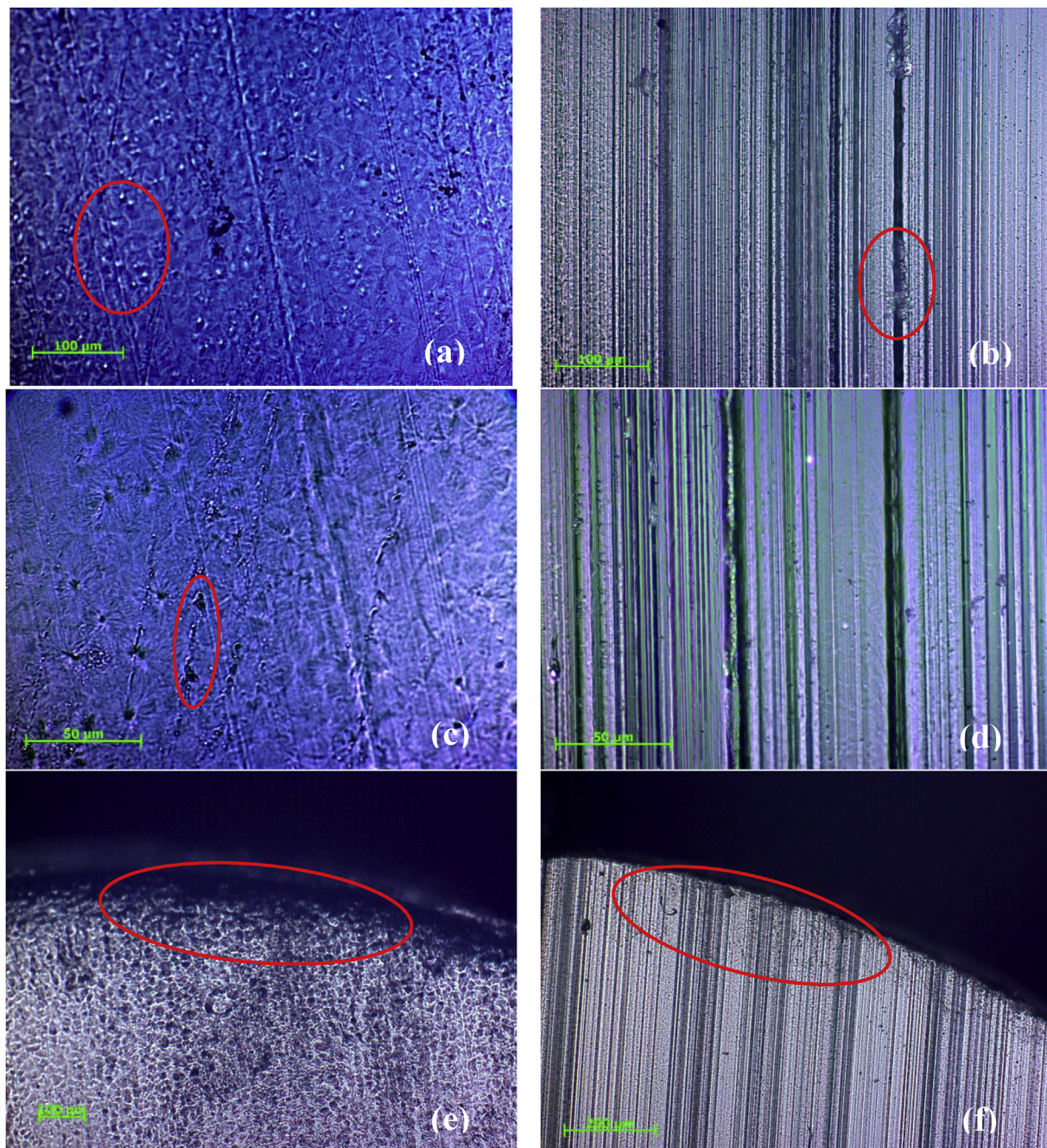


Fig. 9. Microscopic topography analysis of the samples obtained at $p_v = 1.79 \text{ MPa ms}^{-1}$ with $v = 0.22 \text{ m/s}$, dry friction condition: (a, c, and e) worn surfaces of natural polymer; (b,d,f) worn surfaces of composites.

higher magnification revealed the formation of partly melted and scratched surface and re-adhered particles. Further zooming confirms the re-adhered wear debris with lesser dimensions indicating the severe wear regime (Fig. 7e). As reported in literature, PV limit plays an important role in moderate to severe wear regime [25]. Fig. 7 (b, d, e) shows the worn surface of a solid lubricant filled composites at different magnifications. Smooth and slightly re-adhesion of detached particles were observed for the PA 6 filled with the solid lubricant, but the wear loss was moderate. This indicates the coincidence of the friction and wear characteristic curves of polymers and composites at the end of the testing of LPV limit as seen in Fig. 4. So, the effectiveness of solid lubricant in reducing the wear loss was not significant.

Fig. 8 shows the morphological analysis of counter plate against polymer and composites in the wear tests. Fig. 8 (a, c, e) shows the

presence of detached larger size particles and the formation of the discontinued transfer layer over the steel plate against the polymer. The formation of a transfer layer above the metallic surface has been declared in several published works [26–28]. Removal of material from the pin and the occurrence of severe damage due to the uneven asperity contact was ensured. Moreover, the amount of debris outside (Fig. 8e) the wear track guaranteed the decreasing wear resistance of neat polymer when compared with the composite. Similarly, worn surfaces of a counter plate against the composites are shown in Fig. 8 (b, d, f). The decrease in detached particle size and a slight transfer of polymer to steel indicates the functioning of SL at the contact surface. It implies that the wear damage in the case of the SL filled PA 6 polymer composites was not significant, i.e. this composite can perform even at high PV limit. Hence, the addition of SL with the green polymer appears to

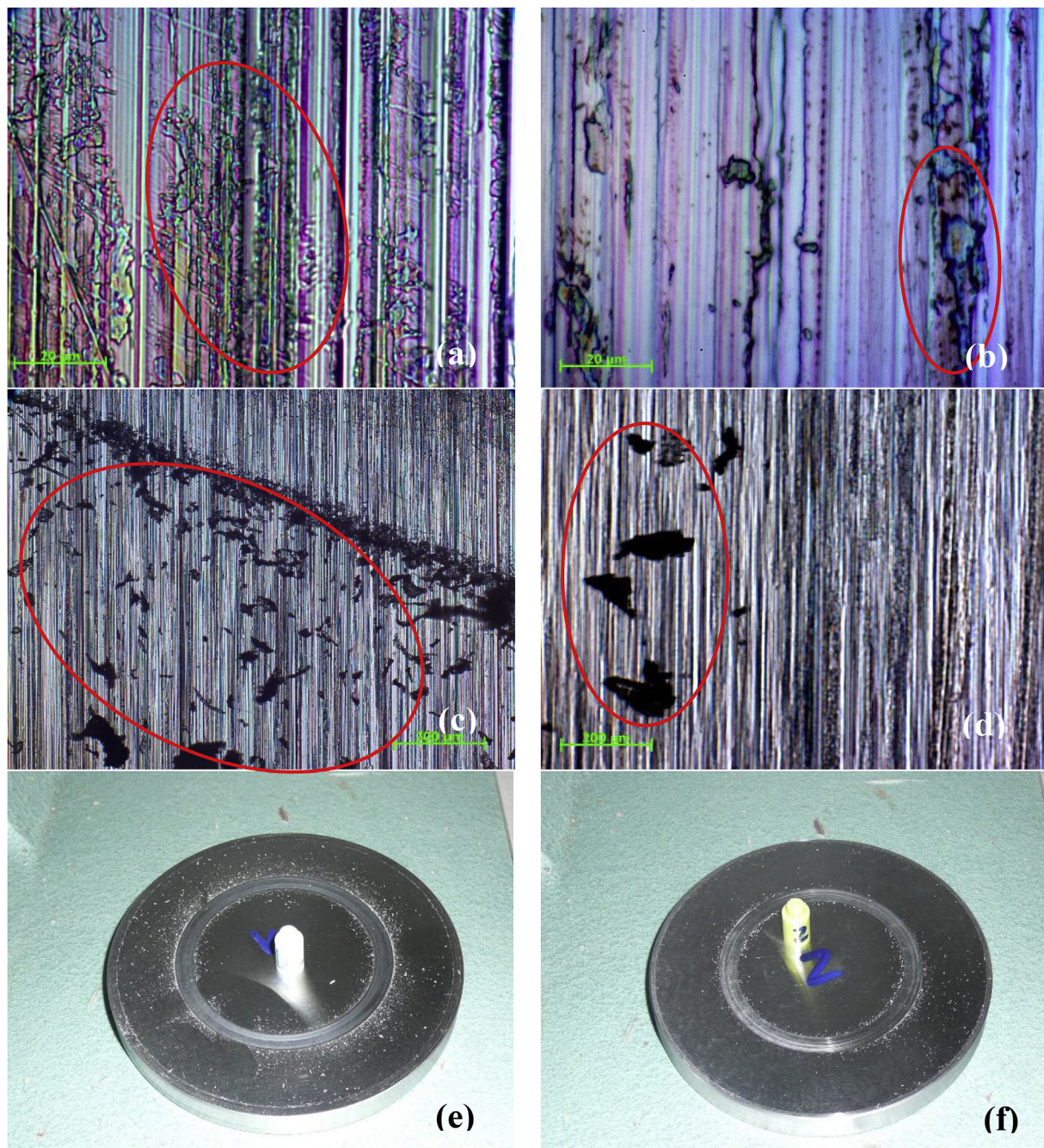


Fig. 10. Microscopic topography analysis of the counter plate obtained at $p_v = 1.79 \text{ MPa ms}^{-1}$ with $v = 0.22 \text{ m/s}$, dry friction condition: (a, c, and e) against with natural polymer; (b,d,f) against with composites.

be more effective in achieving reduced wear damage.

From the microscopy the re-adhesion of transfer layer was evidenced for the neat polymers as observed in the polymer contact surface. This in-turn was validated by the surface investigation of the counterface material. In regards to the transfer layer there are two types 1. Primary layer which is formed between the roughness valleys of the counterface material and 2. Secondary layer which is formed over the roughness peaks as a chunk of flakes. The secondary layer has a tendency to peel off/give away easily when compared with the primary layer. This is explained by the fact that the retaining of debris within the roughness valley plays a significant role in sustaining the primary transfer layer during the sliding process. However, since there are no mechanical constraints from the micro-topography, the polymer layer formed over the roughness peaks has a tendency to get peeled off.

Similar mechanism was also suggested by Sukumaran et al. [17]. Fig. 8c it is evident that a secondary transfer layer plays a major role in increasing friction behaviour of the neat material. Also from literature, it is well known that the characteristics of the primary transfer layer benefit the tribological pair [29,30]. In the current research, the difference between the friction and wear behaviour of the neat and composite material can be attributed to the nature of transfer layer characteristics. A clear indication of primary transfer layer was noted on the counterface tested against the composite material. Particle trapped between the roughness profile is obviously seen in Fig. 8d. However, the neat polymer having the secondary transfer layer did not fully favor the tribological process.

4.2. Worn surface of samples and counter plate at HPV

The worn-out surfaces of the matrix and its composites sample were analyzed through macroscopic analysis to understand the morphological changes arising from the tribological tests. This is illustrated in Fig. 9(a–e). Fig. 9 (a, c, e) exhibited the worn-out surface of PA6G matrix under the HPV condition. Fig. 9a shows the physically deformed and melted surface of samples wherein more debris was found adhered at the edge of the sample. Furthermore, wear tracks with varying intensity of depth were observed. This indicated the filling of gaps by the debris as the result of the increasing contact temperature. Fig. 9c indicates the formation of semi-spherical craters due to the partial melting of the contact surface. A similar crater formation due to the plastic deformation has also been reported by researchers [31]. The higher magnification image is shown in Fig. 9e clearly shows the re-adhered wear debris on the plastically deformed and melted surface. All these wear mechanisms were pointed out in the circle in the microscopic analysis. Fig. 9(b, d, f) shows the morphological analysis of the composite under different magnifications. There was no sign of melting surface in the sample surface. This was explained by the minimal contact temperature (Fig. 9b). A significant amount of the scratched surface with the re-adhered parts is seen in Fig. 9d ploughing action of has a consequence of asperity contact. The micro-ploughing led to the wear resistance of materials getting weaker with high PV limit [32]. Even though the melting was prevalent, no global plastic deformation was evidenced. Fig. 10(a–f) shows the morphology of a steel disc surface against both polymer and composite. The uneven transfer layer from the polymer was seen in the counter plate due to the adhesion process (Fig. 10a). On the other hand, a uniform transfer layer formation was observed in Fig. 10b. Fig. 10c and d clearly shows detached wear debris with varying size besides the wear track. Moreover, a large area of the contact surface was protected by the deposition of polymer films. Fig. 10e helps to visualize the scattered wear debris to the large distance away from the wear track radius, whereas, the scattered distance was minimum for composite (Fig. 10f). The penetration of these wear debris over the material probably led to the high wear rate [33]. This was one of the reasons for getting higher wear loss as seen in Fig. 10f.

In regards to the HPV a similar situation as that of LPV was observed considering the transfer layer formation. However, signs of secondary transfer layer were also observed in the counterface tested against the composite material. This is due to the saturation of the primary transfer layer on the roughness valley. The wear loss in HPV being higher than LPV confirms the fact that more materials have been deposited on the counterface. While comparing the two materials the neat material has generated a relatively significant amount of flaky debris signifying the importance of secondary transfer layer formation.

5. Conclusion

Based on the experimentations and measurements, the following conclusions are drawn:

- The surface energy of the composite (PA6G SL) was a little lower. This influences the adhesive component of friction. In dynamic sliding, there were no significant difference between the neat and composite samples during the running-in friction. However, the composite gives a range of friction coefficients at an early stage of sliding.
- In the model of normal slide bearing application (LPV), with pin-on-disc tests ($p_v = 0.43 \text{ MPa ms}^{-1}$ with $v = 0.14 \text{ m/s}$, dry friction), a slight advantage of composite materials could be observed in friction, wear + deformation and friction heat generation. The difference cannot be considered essential but the additive materials in the composites started acting in the contact zone.
- With an elevated load level (overload condition for the virgin PA6

matrix, HPV), composite samples have proved their advantages. Under $p_v = 1.79 \text{ MPa ms}^{-1}$ with $v = 0.22 \text{ m/s}$, the tribological properties of the composites were significantly better than those of the neat samples.

- The target application based on the experience could be higher pv applications with the high sliding speed for the PA6G SL composite.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compositesb.2018.10.017>.

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